

RIVERSIDE HOSPITAL

Workflow Implementation and Testing Report

Specialist Referral and Consultation Unit

Development of Information Systems Project - UFCFAF-30-3

Report purpose	Project implementation, critical evaluation and testing evidence
Case study	Riverside Hospital Specialist Referral and Consultation Unit
Technology	i*, BPMN, Camunda 8, Python, Git/GitHub
Report date	9 September 2026

Scope of this report

This document summarises the complete Riverside Hospital project journey: assessment interpretation, socio-technical modelling, strategic and operational process modelling, form and variable design, service-task automation, pathway logic, testing, evidence planning and repository preparation. Routine Camunda environment, session and connection troubleshooting is intentionally excluded because it does not contribute to the design or evaluation of the workflow itself.

Contents

1. Executive Summary
 2. Assessment Context and Project Objective
 3. Case Study Analysis and Process Scope
 4. Socio-Technical Modelling with i*
 5. Strategic BPMN Model - AS-IS Process
 6. Operational BPMN Model - Executable TO-BE Workflow
 7. Camunda Forms and Human Interaction Design
 8. Process Variables and Data Alignment
 9. Python Service-Task Automation
 10. Gateway and Conditional Logic
 11. Key Design Issues Identified and Corrected
 12. Testing Strategy and Results
 13. Evidence Capture Strategy
 14. Configuration Management and Repository Work
 15. Critical Evaluation and Limitations
 16. Final Project Position and Submission Readiness
- Appendix A. Core Variables
- Appendix B. Forms and BPMN Task Mapping
- Appendix C. Suggested Expanded Test Matrix

1. Executive Summary

The Riverside Hospital project was developed as a small but coherent information-system prototype for the Specialist Referral and Consultation Unit. The work began with analysis of the assessment specification and the Riverside case study, then moved through i* socio-technical modelling, strategic and operational BPMN, Camunda form design, process-variable definition, Python service-task automation, end-to-end workflow testing and repository preparation.

The central process starts when an external GP Practice submits a referral and ends when the final clinical outcome has been communicated to both the patient and the referring GP. The solution was designed around the Patient Booking Team, Specialist Doctor, Laboratory Team and Medical Secretary, with the GP Practice and Patient treated as external participants. A Workflow Automation System was also modelled in the i* analysis to make the role of automation explicit without incorrectly representing it as a human BPMN lane.

The final operational design supports complete and incomplete referrals, referral clarification, initial appointment scheduling, specialist consultation, optional laboratory investigation, laboratory-result review, discharge, follow-up and treatment outcomes, generation of outcome communications, secretary review and final communication. Five automated service-task roles were designed, exceeding the assessment minimum of three: referral completeness checking, general appointment-reference generation, laboratory appointment-reference generation, specialist notification when laboratory results are available, and draft outcome communication generation.

Testing confirmed the core service workers and a complete no-laboratory discharge pathway. During integration review, several design-level issues were identified, including inconsistent variable names, an unsuitable multi-value field for a Boolean laboratory decision, incomplete separation between laboratory and next-appointment confirmations, and conditional fields in the final outcome form. These were treated as modelling/data-alignment issues rather than platform errors and became important parts of the project evaluation.

Overall outcome

The project demonstrates a clear progression from organisational analysis to executable workflow design. Its strongest features are the explicit separation of human and automated responsibilities, a well-defined process scope, realistic decision points, meaningful service tasks, and documentation that links the case study to the implementation. The remaining quality focus is evidence coverage: each important exception and outcome route should be demonstrated with screenshots before final submission.

2. Assessment Context and Project Objective

The assessment required a single coherent prototype rather than a collection of unrelated diagrams. The work therefore had to show how socio-technical modelling, BPMN modelling, automation, user interaction, software configuration management and testing fit together around one case study.

Assessment requirement	How it was addressed
Strategic Dependency model (i*)	Actors, organisational dependencies, goals, resources, tasks and softgoals were modelled around the Riverside referral process.
Strategic Rationale model (i*)	Internal reasoning was expanded inside actor boundaries using goals, tasks, resources, softgoals, means-end links, task decomposition and contribution links.
Strategic BPMN	A high-level AS-IS business process was modelled to represent the existing manual referral workflow and major hand-offs.
Operational BPMN	A detailed executable workflow was built with user tasks, service tasks, gateways, forms, message interaction and process variables.
Forms / user interfaces	Forms were created for referral entry, clarification, appointments, consultation, laboratory work, outcome decisions and communication review.
At least three automated service tasks	Five automation roles were designed and Python workers were created for the major automated information-processing steps.
Testing	Worker-level and end-to-end process tests were documented with expected results, actual results, pass/fail status, limitations and evidence filenames.

Assessment requirement	How it was addressed
Configuration management	Project artefacts were organised in a Git/GitHub repository with staged commits, documentation, source files and testing evidence.

The work was intentionally developed in stages. Early discussions focused on understanding what the assessment meant, distinguishing i* from BPMN, deciding which actors belonged in which model, and defining a realistic project sequence. Later work became more implementation-oriented: form fields, variables, gateway expressions, Python worker inputs and outputs, test pathways and evidence capture were checked against one another.

3. Case Study Analysis and Process Scope

The case study describes a specialist outpatient unit that receives referrals from independent GP practices. The existing process relies heavily on email, shared documents, manual checks and separate administrative systems. This creates uncertainty about referral status, duplicated effort, inconsistent communication and delays when information is missing or when staff are waiting for laboratory results or consultation notes.

The project scope was therefore defined around one clear start and end point. The process starts when Riverside Hospital receives a GP referral and ends when the final clinical outcome has been communicated to the patient and the referring GP. Emergency care, inpatient treatment and real integrations with NHS, GP, laboratory or email systems were excluded from the prototype.

Participant	Role in the model
GP Practice	External participant that sends the referral, provides clarification when required and receives the final outcome.
Patient	External participant that receives appointment information and the final outcome communication.
Patient Booking Team	Captures/refines referral information, manages clarification and schedules initial, laboratory, follow-up and treatment appointments.
Specialist Doctor	Conducts consultation, records clinical information, determines whether laboratory testing is required, reviews results and confirms the final outcome.
Laboratory Team	Performs laboratory investigation and records/uploads results.
Medical Secretary	Reviews and finalises generated communications and confirms the outcome has been sent.
Workflow Automation System	Performs background checks, reference generation, result notification and draft communication generation.

The high-level workflow was stabilised as follows:

1. Receive and review the GP referral.
2. Automatically evaluate referral completeness and identify missing information.
3. If incomplete, request clarification from the GP, update the referral and repeat validation.
4. Schedule the initial specialist consultation and generate the appointment reference/confirmation.

5. Conduct the specialist consultation and capture notes, outcome summary and the laboratory decision.
6. If laboratory work is required, schedule the laboratory appointment, perform the investigation, upload the result and return it to the Specialist Doctor for review.
7. Confirm the final outcome as discharge, follow-up or treatment.
8. Where follow-up or treatment is required, schedule the next appointment and generate its confirmation.
9. Generate draft outcome communication, have the Medical Secretary review/finalise it, and communicate the final outcome to both patient and GP.

4. Socio-Technical Modelling with i*

4.1 Strategic Dependency model

The Strategic Dependency (SD) model was used to show who depends on whom and for what. This was deliberately kept separate from BPMN. BPMN describes process flow, while the i* SD model describes organisational dependency and intentional relationships.

A recurring design question was whether the workflow system should appear as an actor. The final modelling decision was to include Workflow Automation in the i* analysis because the hospital staff depend on automated capabilities such as completeness checking, appointment-reference generation and outcome drafting. In contrast, it was not treated as a human lane in BPMN; the executable workflow expresses automation through service tasks and the process engine.

Key dependency area	Examples modelled
Referral information	GP Practice provides referral information to the Patient Booking Team; the booking team depends on the GP for clarification when information is missing.
Appointment arrangement	Patient depends on the Patient Booking Team for consultation/laboratory/next-appointment details; the booking team depends on automation for consistent references and messages.
Clinical assessment	The Patient Booking Team and Patient depend on the Specialist Doctor for clinical assessment and next-step decisions.
Laboratory pathway	Specialist Doctor depends on the Laboratory Team for investigation/results; the doctor also depends on automation for timely results-available notification.
Outcome communication	Medical Secretary depends on clinical notes/outcome information and generated drafts; Patient and GP depend on the hospital for accurate, timely final communication.

The SD work also required repeated checks on dependency direction. The depender is the actor that needs something; the dependee is the actor expected to provide or achieve it. Keeping this distinction consistent prevented arrows from being drawn according to visual convenience rather than organisational meaning.

4.2 Strategic Rationale model

The Strategic Rationale (SR) model expanded the reasoning inside each actor boundary. The model used the standard i* element types: hard goals, softgoals, tasks and resources. Tasks were linked to goals using means-end reasoning, complex work was broken down through task decomposition, and contribution links were used where a task positively or negatively affected a softgoal.

Actor boundary	Main rationale represented
Patient Booking Team	Referral readiness, complete referral acceptance, appointment scheduling, clarification handling, patient communication and administrative quality.
GP Practice	Preparing complete referral information, supplying clarification, receiving the final outcome and supporting accurate/timely information exchange.
Specialist Doctor	Clinical assessment, consultation notes, laboratory decision, results review, final clinical outcome and quality of clinical decision-making.
Laboratory Team	Completion of requested investigations and production of reliable laboratory results.
Medical Secretary	Reviewing/finalising communications and ensuring the patient and GP receive a consistent final outcome.
Patient	Receiving appointments, attending the process and receiving understandable outcome information.
Workflow Automation	Completeness checking, reference generation, notifications and draft communication support.

An important notation issue was the treatment of resources. For example, Consultation Notes are not themselves a goal and should not be forced into a means-end relationship. They are resources produced/used by tasks such as Record Consultation Notes and Determine Clinical Outcome. This distinction helped keep the SR model conceptually aligned with i* rather than turning it into a second BPMN diagram.

5. Strategic BPMN Model - AS-IS Process

The strategic BPMN represents the client's existing process at a high level. It was intentionally non-executable and focused on the major business hand-offs rather than form bindings or service-task implementation. The GP Practice and Patient were represented as external participants, while Riverside Hospital was divided into the four main operational lanes: Patient Booking Team, Specialist Doctor, Laboratory Team and Medical Secretary.

The strategic model contains the referral review, clarification loop, appointment arrangement, specialist consultation, laboratory decision, optional investigation/results activity, final outcome and outcome communication. Message flows were used where information crosses organisational boundaries, particularly between Riverside Hospital and the GP Practice or Patient.

Why the strategic model mattered

Building the AS-IS model before the executable version created a clean baseline. It made it easier

to justify which manual activities should remain user tasks and which repetitive information-processing activities were appropriate candidates for automation.

6. Operational BPMN Model - Executable TO-BE Workflow

The operational BPMN converted the case study into a detailed executable design. Unlike the strategic model, this version needed explicit user tasks, gateways, forms, service tasks, message interaction and process variables. The process was developed section by section so that each part could be checked against the case study and later mapped to a form or Python worker.

Operational section	Main BPMN behaviour
Referral intake	Message start for GP referral, referral review/entry and automated completeness checking.
Clarification loop	Incomplete referrals branch to clarification request/receipt, referral update and re-check.
Initial consultation	Booking team schedules consultation, automation creates reference/confirmation, patient is informed.
Consultation	Specialist conducts consultation and records notes/outcome, including the laboratory decision.
Laboratory branch	If required: schedule lab appointment, generate lab reference/confirmation, inform patient, perform investigation, upload results, notify specialist and review results.
Final outcome	Specialist confirms the final outcome and selects discharge, follow-up or treatment.
Next appointment	Follow-up/treatment routes schedule a next appointment, generate a reference/confirmation and inform the patient.
Outcome communication	All routes converge; draft outcome communication is generated, reviewed/finalised and sent to patient and GP.

The final outcome section was particularly important because the workflow had to model three distinct clinical outcomes without duplicating the entire communication process. The selected approach uses one gateway controlled by `finalOutcomeType`. Discharge proceeds directly toward outcome communication, while follow-up and treatment create a next appointment before converging back into the common outcome-letter pathway.

7. Camunda Forms and Human Interaction Design

Forms were used for human activities so that the process could collect and display the variables required by later gateways and service tasks. The form structure mirrors the operational BPMN rather than being designed as a separate interface layer.

Form	BPMN use
referral-details.form	Enter / Review Referral Details; Update Referral Details
referral-clarification.form	Request Referral Clarification

Form	BPMN use
referral-clarification-received.form	Receive Referral Clarification
initial-consultation-appointment.form	Schedule / Confirm Initial Consultation
appointment-confirmation.form	Inform Patient of Appointment
consultation-completion.form	Conduct Specialist Consultation
consultation-outcome.form	Record Consultation Notes & Outcome
laboratory-appointment.form	Schedule Laboratory Appointment
lab-appointment-confirmation.form	Inform Patient of Lab Appointment
laboratory-investigation-completion.form	Perform Laboratory Investigation
laboratory-result.form	Upload Laboratory Results
laboratory-results-review.form	Review Laboratory Results
final-clinical-outcome.form	Confirm Final Clinical Outcome
next-appointment.form	Schedule Follow-Up Consultation; Schedule Treatment Appointment
next-appointment-confirmation.form	Inform Patient of Next Appointment
outcome-communication-review.form	Review / Finalise Outcome Communication
outcome-send-confirmation.form	Send Final Outcome to Patient and GP

Several form decisions were refined during integration. The laboratory-required decision was changed from a Checkbox Group to a single Boolean checkbox because the gateway needs true/false rather than a list. The final clinical outcome form was also adjusted so that only the relevant detail field is shown for the selected outcome: follow-up requirements for follow-up, treatment plan for treatment and discharge advice for discharge. A dedicated next-appointment confirmation form was separated from the laboratory confirmation form so that the two concepts do not share unrelated variables.

8. Process Variables and Data Alignment

A major part of the implementation work was making the BPMN, forms and Python workers speak the same variable language. The process-scope-and-variables document became the reference point for the intended data contract. Variables were grouped by business purpose rather than left as ad-hoc form field names.

Variable group	Representative variables
Referral information	patientName, nhsNumber, dateOfBirth, contactDetails, referringGP, gpPractice, referralReason, referralPriority
Referral validation	referralComplete, missingReferralFields
Clarification	clarificationRequest, clarificationReceived
Initial appointment	consultationDate, consultationTime, specialistDoctor, appointmentReference, appointmentConfirmation

Variable group	Representative variables
Consultation	consultationCompleted, consultationNotes, consultationOutcomeSummary, labRequired, laboratoryTestType, laboratoryTestReason
Laboratory appointment	labAppointmentDate, labAppointmentTime, labAppointmentReference, labAppointmentConfirmation
Laboratory results	laboratoryInvestigationCompleted, laboratoryResultDate, laboratoryResult, labResultAvailable, specialistResultNotification, laboratoryReviewNotes
Final outcome	finalClinicalOutcomeSummary, finalOutcomeType, clinicalPlan, followUpRequirements, treatmentPlan, dischargeAdvice
Next appointment	nextAppointmentDate, nextAppointmentTime, nextAppointmentNotes, nextAppointmentReference, nextAppointmentConfirmation
Outcome communication	draftPatientOutcomeLetter, draftGPOutcomeLetter, finalPatientOutcomeCommunication, finalGPOutcomeCommunication, outcomeCommunicationConfirmed

The most important variable-alignment decisions were not cosmetic. They determined whether gateway expressions could evaluate correctly and whether a value generated by one task could actually be displayed or reviewed by the next task. Meaningful names were preferred over component-generated identifiers because the former make the BPMN, worker code and evidence understandable during assessment.

8.1 Preferred naming standards established during review

Area	Inconsistency identified	Preferred standard / rationale
Missing referral fields	missingFields vs missingReferralFields	Use missingReferralFields. It clearly describes a list produced by the referral completeness worker and matches the documented process variable.
Laboratory reason	laboratoryTestReason vs a generated text-area key	Use laboratoryTestReason. It is meaningful and stable across the form, BPMN and worker code.
Laboratory confirmation	labAppointmentConfirmationMessage vs labAppointmentConfirmation	Use one documented output consistently; labAppointmentConfirmation is the cleaner process-level name.
Laboratory result notification	specialistResultsAvailable / specialistResultsNotification vs labResultAvailable / specialistResultNotification	Choose one pair and update both forms and workers. The documented singular names are clearer and avoid duplicated concepts.
Outcome drafts	draftOutcomeLetter or final communication variables vs draftPatientOutcomeLetter / draftGPOutcomeLetter	Generate draft variables first; finalPatientOutcomeCommunication and finalGPOutcomeCommunication should represent secretary-reviewed content.

Area	Inconsistency identified	Preferred standard / rationale
Appointment outputs	appointmentReference only vs separate initial/next outputs	Keep appointmentReference/appointmentConfirmation for the initial consultation and nextAppointmentReference/nextAppointmentConfirmation for follow-up or treatment.

Current-code verification note

The uploaded code snapshot used during this report review still contains some of the older output names in the laboratory and outcome-letter workers. The report therefore distinguishes between the preferred data contract and what could be verified in that snapshot. Before final submission, the BPMN, forms, process-scope document and Python return dictionaries should all be checked against the same final variable names.

9. Python Service-Task Automation

The workflow automation was implemented with Python workers using the Camunda Python SDK. Each worker subscribes to a job type, reads process variables, performs a small realistic information-processing function and returns process variables to the workflow.

Worker / job type	Purpose	Key outputs
check_referral_completeness.py / check-referral-completeness	Checks the eight required referral fields and identifies missing information.	referralComplete, missingReferralFields, referralCheckMessage
generate_appointment_reference.py / generate-appointment-reference	Creates a unique Riverside appointment reference and a confirmation message.	appointmentReference, appointmentConfirmation
generate_lab_appointment_reference.py / generate-lab-appointment-reference	Creates a laboratory appointment reference and patient-facing confirmation details.	labAppointmentReference plus laboratory confirmation output
notify_specialist_results_available.py / notify-specialist-results-available	Checks that a laboratory result exists and produces a notification for specialist review.	results-available Boolean and notification text
generate_draft_outcome_letter.py / generate-draft-outcome-letter	Builds a draft clinical outcome communication from patient and clinical outcome information.	draft/final communication variables depending on the current implementation binding

A separate start script was also created to publish the GP referral start message for an end-to-end test scenario. This made it possible to trigger the process in a repeatable way instead of relying only on manual process starts.

The service tasks were selected because they are plausible automation candidates. They do not replace clinical judgement. Instead, they reduce repetitive checking, produce consistent administrative references, notify staff when results are available and prepare draft text for human review. This is closely aligned with the case study requirement that automation should support staff rather than make clinical decisions on their behalf.

10. Gateway and Conditional Logic

The operational workflow contains three major decision points. Each was standardised around a small, predictable variable domain.

Gateway	Controlling variable	Expected values	Purpose
Referral Complete?	referralComplete	true / false	Either continue to booking or enter the clarification loop.
Laboratory Test Required?	labRequired	true / false	Either proceed directly toward final outcome or enter the laboratory pathway.
Final Outcome Type?	finalOutcomeType	discharge / followUp / treatment	Selects discharge, follow-up or treatment behaviour before the common outcome communication stage.

The final clinical outcome form uses conditional visibility to keep the interface relevant to the selected route. When finalOutcomeType is followUp, followUpRequirements should appear. When it is treatment, treatmentPlan should appear. When it is discharge, dischargeAdvice should appear. These conditions improve data quality because the doctor is not asked to complete unrelated fields.

The laboratory decision was another important example of matching form behaviour to BPMN semantics. A multi-select component can legitimately return an array, but the gateway was designed for a Boolean. The form was therefore simplified so that labRequired is a true/false value, which is easier to reason about and test.

11. Key Design Issues Identified and Corrected

The most valuable problems found during the project were not runtime problems; they were design and integration issues that affected whether the model meant what it was supposed to mean. The table below records the main corrections because these demonstrate the project's iterative engineering process.

Issue	Why it mattered	Resolution / design decision
Workflow Automation missing from early i* thinking	Automation responsibilities risked being invisible in the socio-technical analysis.	Workflow Automation was explicitly included as an i* actor with dependencies for checking, reference generation, notification and drafting. It remains absent as a human BPMN lane.
Confusion between i* link types	Tasks, goals and resources could be connected with semantically incorrect links.	Means-end was reserved for tasks that achieve goals, decomposition for task parts, contribution for softgoals, and resources were treated as things produced/needed by tasks.
Strategic vs operational BPMN distinction	The high-level model risked becoming too detailed or executable.	The AS-IS model was kept strategic; executable details such as forms, variables and service tasks were moved into the operational model.
Laboratory route initially under-specified	The process did not fully demonstrate the required lab appointment and results-notification automation.	Dedicated laboratory appointment-reference and specialist-results-notification service tasks were added to the design and worker set.

Issue	Why it mattered	Resolution / design decision
labRequired data type mismatch	A Checkbox Group produced a collection rather than a Boolean expected by the gateway.	The control was changed to a single Boolean checkbox so the gateway can reliably test true/false.
Inconsistent form/worker variable names	Generated values could be present but unavailable to downstream tasks because the key names differed.	A process-wide naming standard was documented and mismatches were reviewed across forms, workers and BPMN.
Final outcome fields all visible / incorrect logic	The final clinical outcome form could collect irrelevant or confusing information.	Conditional visibility was defined for follow-up requirements, treatment plan and discharge advice based on finalOutcomeType.
Next appointment reused laboratory confirmation form	Follow-up/treatment communication risked carrying laboratory-specific variables and wording.	A dedicated next-appointment-confirmation.form was defined for INFORM PATIENT OF NEXT APPOINTMENT.
Outcome draft/final variables blurred	Automatically generated text could be treated as final before secretary review.	The intended data model separates draft patient/GP letters from final reviewed communications.
Testing evidence too narrow	A single happy-path execution would not adequately demonstrate exception behaviour or all final outcomes.	The evidence plan was expanded to include incomplete referral, laboratory, follow-up and treatment routes in addition to the complete no-lab discharge path.

12. Testing Strategy and Results

12.1 Testing objective and environment

Testing was designed to confirm that the workflow behaviour matched the case study and that the major technical components could exchange process data correctly. The test scope covered automated service tasks, Python job workers, forms, gateway decisions, process variables and end-to-end execution. All test data was fictional.

Environment item	Used in testing
Camunda	Camunda 8 local workflow environment
Modeler	Camunda Modeler 5.49.0
Operational tools	Operate and Tasklist
Automation	Python and Camunda Python SDK
Development environment	Visual Studio Code
Source control	Git / GitHub repository

12.2 Confirmed test cases from the current test report

ID	Scenario	Expected / actual result	Status
TC01	Check referral completeness service task	Worker should pick up the job and return a completeness result. It was picked up successfully and returned referralComplete = true in the isolated test.	PASS

ID	Scenario	Expected / actual result	Status
TC02	Generate appointment reference and confirmation	Worker should generate a unique appointment reference and confirmation message. It completed successfully and returned both values.	PASS
TC03	Generate draft outcome letter	Worker should generate draft outcome content from patient name, outcome type and consultation notes. It completed successfully and returned generated communication text.	PASS
TC04	End-to-end complete referral -> no laboratory -> discharge	Complete referral should pass validation, schedule consultation, follow the no-lab route, select discharge, generate/review outcome communication and reach process completion. The recorded test completed successfully.	PASS

TC01 contained an important limitation in its earliest isolated form: the completeness result was temporarily fixed to true. The final referral worker was later revised to inspect the actual eight referral fields and construct missingReferralFields. This makes the worker meaningful for both the happy path and the incomplete-referral scenario.

TC04 is the strongest confirmed integration test in the supplied report because it exercised the process from referral receipt through final communication. It showed that the complete-referral branch, appointment generation, consultation, no-laboratory branch, discharge outcome and outcome communication could work together in one process instance.

Known evidence filename / evidence group	Purpose
TC01_check_referral_completeness_camunda.png	Camunda-side evidence that the completeness service task executed.
TC01_check_referral_completeness_worker.png	Worker console evidence showing the job was picked up and variables returned.
TC02_generate_appointment_reference_camunda.png	Camunda-side evidence of appointment-reference automation.
TC02_generate_appointment_reference_worker.png	Worker output showing the generated reference and confirmation.
TC03_generate_draft_outcome_letter_camunda.png	Camunda-side evidence of outcome-letter generation.
TC03_generate_draft_outcome_letter_worker.png	Worker output showing generated communication text.
TC04 evidence set	Final outcome, process history, completed-process and final process-variable screenshots stored under testing/evidence.

Evidence status

The report marks only the scenarios that were explicitly recorded as successful in the supplied test report as PASS. The broader feedback requested additional end-to-end evidence for

incomplete referral, laboratory, follow-up and treatment routes. The workflow design now contains those routes, but they should be marked PASS in a final test matrix only after the corresponding process instances and screenshots have actually been captured.

12.3 Expanded pathway testing required for strong assessment evidence

Scenario	What should be demonstrated	Evidence focus
Complete referral -> no lab -> discharge	Current TC04 baseline.	Referral variables, gateway path, generated appointment, final outcome, generated communication and completed process.
Incomplete referral -> clarification -> corrected referral	Automated validation fails, missing items are identified, clarification is received, referral is updated and validation succeeds on re-check.	Tasklist forms before/after correction, missingReferralFields, gateway route and re-check.
Complete referral -> lab required -> result -> review -> outcome	Boolean lab branch, lab appointment, lab worker, result upload, specialist notification/review and return to final outcome.	Tasklist lab forms, worker output, Operate path and lab-related variables.
Follow-up outcome	finalOutcomeType = followUp leads to follow-up scheduling and patient next-appointment confirmation before outcome communication.	Conditional form field, next appointment variables, gateway route and confirmation form.
Treatment outcome	finalOutcomeType = treatment leads to treatment scheduling and patient next-appointment confirmation before outcome communication.	Treatment-plan field, next appointment variables, gateway route and completed communication.

13. Evidence Capture Strategy

The assessment requires evidence that forms, gateways and service tasks have been tested. The evidence should therefore show both the user-facing inputs and the engine-side consequences. A screenshot of only one tool is rarely enough to explain the full behaviour.

Evidence source	What it proves	Best use
Tasklist	What a human user saw and submitted through a form.	Referral input, clarification, appointment scheduling, consultation data, lab result entry, final outcome selection and secretary review.
Operate	How the process instance moved through the BPMN and what process variables existed.	Gateway route, active/completed tasks, full process history, final variable state and successful completion.
Python worker terminal	That the service task was actually picked up and what values the worker read/returned.	Referral completeness, generated references, specialist notification and generated outcome text.

For each significant test, the most persuasive evidence set is therefore: a Tasklist screenshot for the important human input, an Operate screenshot showing the selected route, an Operate variables screenshot where the gateway/service output matters, and a worker-console screenshot for any

automated service task under test. Not every single task needs all four; screenshots should be selected to prove the behaviour being claimed.

Evidence filenames should remain test-case oriented and descriptive. This makes the testing folder readable without opening every image and supports a clear link between the test report and repository evidence.

14. Configuration Management and Repository Work

Git and GitHub were used throughout the project so that the modelling, automation and evidence could be reviewed as one project rather than as scattered local files. Repository work included initial setup, staging and committing models, adding form files and worker source code, storing documentation, and preparing testing evidence.

The assessment specifically requires the i* models to be available in an accepted submission format. The source draw.io files remain useful because they are editable, but the final repository should also contain exported PDF or DOCX versions and the required text representation where specified by the assessment. The same principle applies to the final testing report: this PDF can replace the earlier reporttest.md as the human-readable submitted report, while source markdown may still be retained if desired for version history.

Repository artefact area	Contents / purpose
models	Strategic Dependency, Strategic Rationale, strategic BPMN and operational BPMN source/export files.
forms	Camunda form files mapped to user tasks.
workers / automation	Python service-task implementations and start/test helper script.
documentation	README, process-scope-and-variables.md and supporting explanations.
testing	Test report, scenario records and evidence screenshots.
version history	Meaningful commits showing staged development rather than one final bulk upload.

Meaningful commit messages were used at project milestones, for example after completing the Strategic Rationale model, after completing operational workflow sections, after correcting automation/data alignment and for the final repository clean-up. This demonstrates software configuration-management practice rather than using Git only as a submission folder.

15. Critical Evaluation and Limitations

15.1 Strengths of the solution

- The process boundary is clear and consistently tied to the Riverside Hospital case study.
- Human responsibilities and automated responsibilities are separated rather than mixed together.
- The operational workflow contains realistic exception behaviour, especially the incomplete-referral loop and optional laboratory pathway.
- Automation is used for information-processing tasks rather than clinical judgement.

- The final outcome design avoids unnecessary duplication by converging discharge, follow-up and treatment routes before communication.
- The process-scope-and-variables document provides a useful data contract for forms, gateways and workers.
- The project uses version control and test evidence, which supports reproducibility and assessment review.

15.2 Limitations

- The implementation is a local prototype and does not integrate with real NHS, GP, laboratory, email or patient-notification systems.
- All patient and GP information used for testing is fictional.
- The generated communications are draft text and are not intended to replace clinical or administrative review.
- The appointment and notification workers simulate business-system behaviour; they do not reserve real hospital resources or deliver real messages.
- The most recent uploaded code snapshot still shows some variable-name inconsistencies compared with the documented data contract. A final consistency pass is necessary before submission.
- The current confirmed test report proves the core workers and the complete no-lab discharge route. Stronger assessment evidence requires captured executions of incomplete-referral, laboratory, follow-up and treatment scenarios.

15.3 Reflection on the development process

The project improved as the models were forced to agree with each other. Several early questions that appeared to be drawing questions - arrow direction, whether a resource should use a means-end link, whether Workflow Automation should be a lane, or whether a checkbox should be single or multi-select - turned out to affect the correctness of the later implementation. Treating the SD/SR models, BPMN, forms, variables and workers as one connected design was therefore more valuable than perfecting any one artefact in isolation.

The integration review was also useful because it exposed a common workflow-automation problem: two components can both be individually correct and still fail to work together if they use different variable names or different data types. Establishing process-level naming standards and evidence-driven testing was the main step that moved the project from a set of functioning pieces toward a coherent prototype.

16. Final Project Position and Submission Readiness

The Riverside Hospital project now has the main artefacts required by the assessment: i* socio-technical models, strategic and operational BPMN, an automated Camunda workflow design, user forms, more than the minimum number of service-task implementations, process documentation, test cases and repository-based configuration management.

Before final submission, the most important quality check is consistency rather than adding more features. The final BPMN, form keys, Python return dictionaries and process-scope-and-variables document should use exactly the same variable names. The evidence folder should then contain screenshots for the remaining key exception/outcome routes so that every PASS result in the test report is supported by visible proof.

Recommended final submission check

1) Verify final variable names across BPMN, forms and all five workers. 2) Run and capture the incomplete-referral, laboratory, follow-up and treatment pathways. 3) Export the final SD/SR models into the formats required by the assessment while keeping draw.io sources. 4) Replace reporttest.md in the submission-facing testing folder with this PDF (or retain the markdown only as a source copy). 5) Make one final meaningful Git commit and confirm the repository is accessible to the module team.

Once those evidence and consistency checks are complete, the project will present a clear story: the Riverside case study was analysed, organisational dependencies were modelled, the manual process was understood, a detailed operational design was created, realistic automation was implemented, the process was tested, defects were evaluated and the work was managed through version control.

Appendix A. Core Variables

Group	Preferred process variables
Referral	patientName; nhsNumber; dateOfBirth; contactDetails; referringGP; gpPractice; referralReason; referralPriority
Validation	referralComplete; missingReferralFields
Clarification	clarificationRequest; clarificationReceived
Initial appointment	consultationDate; consultationTime; specialistDoctor; appointmentReference; appointmentConfirmation
Consultation	consultationCompleted; consultationNotes; consultationOutcomeSummary; labRequired; laboratoryTestType; laboratoryTestReason
Lab appointment	labAppointmentDate; labAppointmentTime; labAppointmentReference; labAppointmentConfirmation
Lab result/review	laboratoryInvestigationCompleted; laboratoryResultDate; laboratoryResult; labResultAvailable; specialistResultNotification; laboratoryReviewNotes
Final outcome	finalClinicalOutcomeSummary; finalOutcomeType; clinicalPlan; followUpRequirements; treatmentPlan; dischargeAdvice
Next appointment	nextAppointmentDate; nextAppointmentTime; nextAppointmentNotes; nextAppointmentReference; nextAppointmentConfirmation
Communication	draftPatientOutcomeLetter; draftGPOutcomeLetter; finalPatientOutcomeCommunication; finalGPOutcomeCommunication; outcomeCommunicationConfirmed

Appendix B. Forms and BPMN Task Mapping

Form	Task(s)
referral-details.form	Enter / Review Referral Details; Update Referral Details
referral-clarification.form	Request Referral Clarification
referral-clarification-received.form	Receive Referral Clarification
initial-consultation-appointment.form	Schedule / Confirm Initial Consultation
appointment-confirmation.form	Inform Patient of Appointment
consultation-completion.form	Conduct Specialist Consultation
consultation-outcome.form	Record Consultation Notes & Outcome
laboratory-appointment.form	Schedule Laboratory Appointment
lab-appointment-confirmation.form	Inform Patient of Lab Appointment
laboratory-investigation-completion.form	Perform Laboratory Investigation
laboratory-result.form	Upload Laboratory Results
laboratory-results-review.form	Review Laboratory Results

Form	Task(s)
final-clinical-outcome.form	Confirm Final Clinical Outcome
next-appointment.form	Schedule Follow-Up Consultation; Schedule Treatment Appointment
next-appointment-confirmation.form	Inform Patient of Next Appointment
outcome-communication-review.form	Review / Finalise Outcome Communication
outcome-send-confirmation.form	Send Final Outcome to Patient and GP

Appendix C. Suggested Expanded Test Matrix

ID	Scenario	Status in this report	Minimum evidence before marking PASS
TC01	Referral completeness worker	PASS	Camunda task + worker output
TC02	Appointment reference worker	PASS	Camunda task + worker output
TC03	Draft outcome worker	PASS	Camunda task + worker output
TC04	Complete referral -> no lab -> discharge	PASS	Process path + final variables + completion evidence
TC05	Incomplete referral -> clarification -> corrected referral	To be evidenced	Tasklist before/after correction + missing fields + re-check path
TC06	Complete referral -> laboratory route -> result review	To be evidenced	Lab forms + lab workers + Operate path + result/review variables
TC07	Follow-up final outcome	To be evidenced	Outcome form + follow-up route + next appointment confirmation
TC08	Treatment final outcome	To be evidenced	Outcome form + treatment route + next appointment confirmation

Document Basis

This report was prepared from the Riverside Hospital case study and assessment specification, the project process-scope-and-variables document, the current Python worker set, the project feedback document, and the testing report content supplied for TC01-TC04. Routine Camunda connection/session troubleshooting has been excluded by design.

Key project files reviewed include: process-scope-and-variables.md; check_referral_completeness.py; generate_appointment_reference.py; generate_lab_appointment_reference.py; notify_specialist_results_available.py; generate_draft_outcome_letter.py; start_riverside_process.py; and the assessment feedback document dated 9 September 2026.